



NICIS

NATIONAL INTEGRATED
CYBERINFRASTRUCTURE SYSTEM

DIRISA

Introduction to Python

Elias Sepuru (Senior Software Engineer) | 05 July 2026

AN INITIATIVE OF:



science, technology
& innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



CSIR
Touching lives through innovation

80th
anniversary



Topics to be discussed

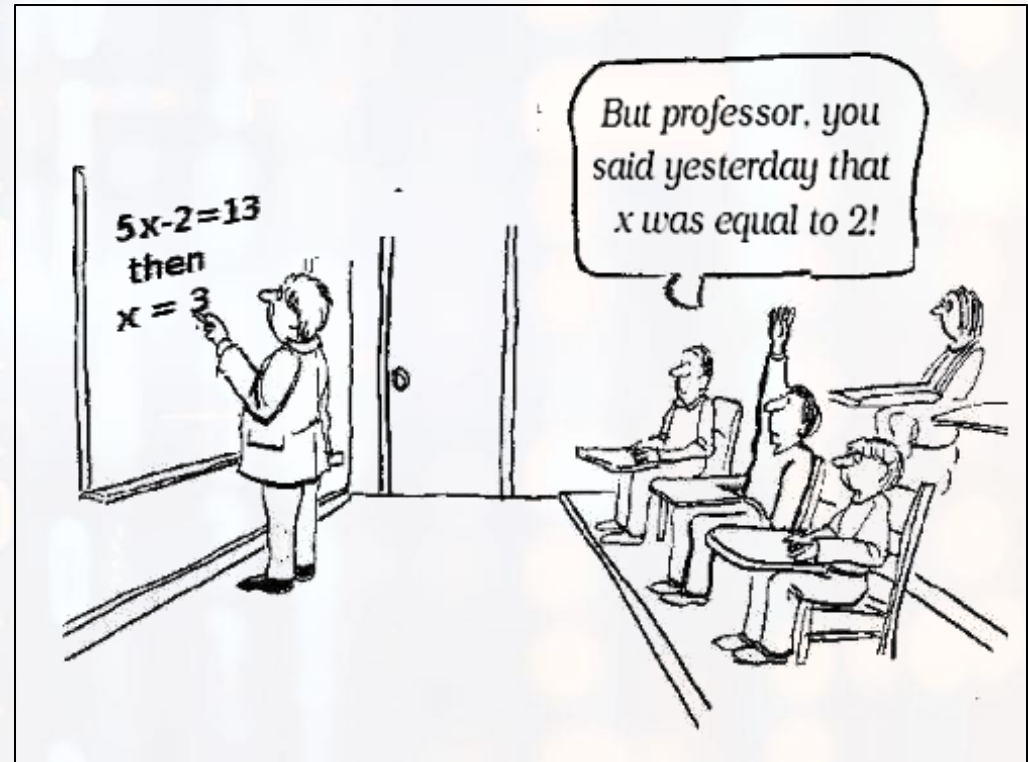
- Variables, Data Types and Print Statements
- Functions
- Lists and Dictionaries
- If and For Statement
- Loop Statements



Variables

- Variables are containers for storing data values - so that a computer can use it later.

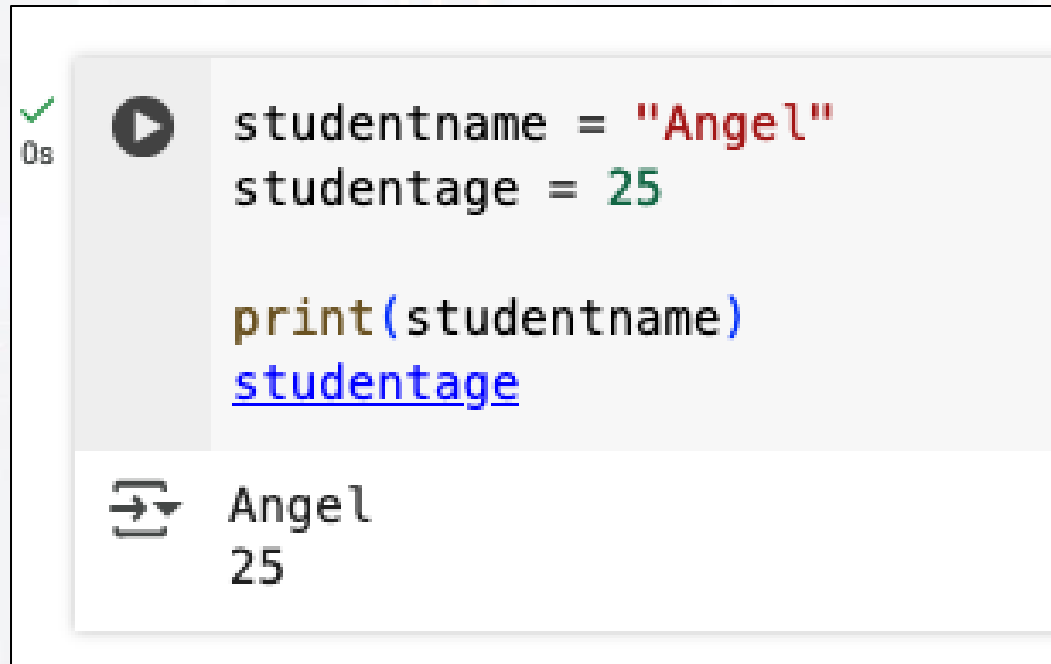
```
name = "Thabo"
```





Print Statement

- You can display or print something using the 'Print Statement' or without using it.



```
✓  
0s ▶ studentname = "Angel"  
    studentage = 25  
  
    print(studentname)  
    studentage  
⇒ Angel  
   25
```

The image shows a snippet of Python code being executed. It consists of two lines of assignment: `studentname = "Angel"` and `studentage = 25`. This is followed by a `print` statement that prints the value of `studentname`. Below the code, the output is displayed as `Angel` and `25` on separate lines. A green checkmark and '0s' indicate successful execution.



Data Types

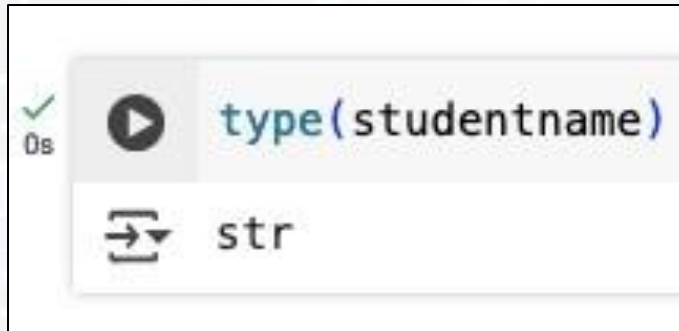
Data Type	Size	Description
byte	1 byte	Stores whole numbers from -128 to 127
short	2 bytes	Stores whole numbers from -32,768 to 32,767
int	4 bytes	Stores whole numbers from -2,147,483,648 to 2,147,483,647
long	8 bytes	Stores whole numbers from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4 bytes	Stores fractional numbers. Sufficient for storing 6 to 7 decimal digits
double	8 bytes	Stores fractional numbers. Sufficient for storing 15 decimal digits
boolean	1 bit	Stores true or false values
char	2 bytes	Stores a single character/letter or ASCII values



Data Types

To check variable type:

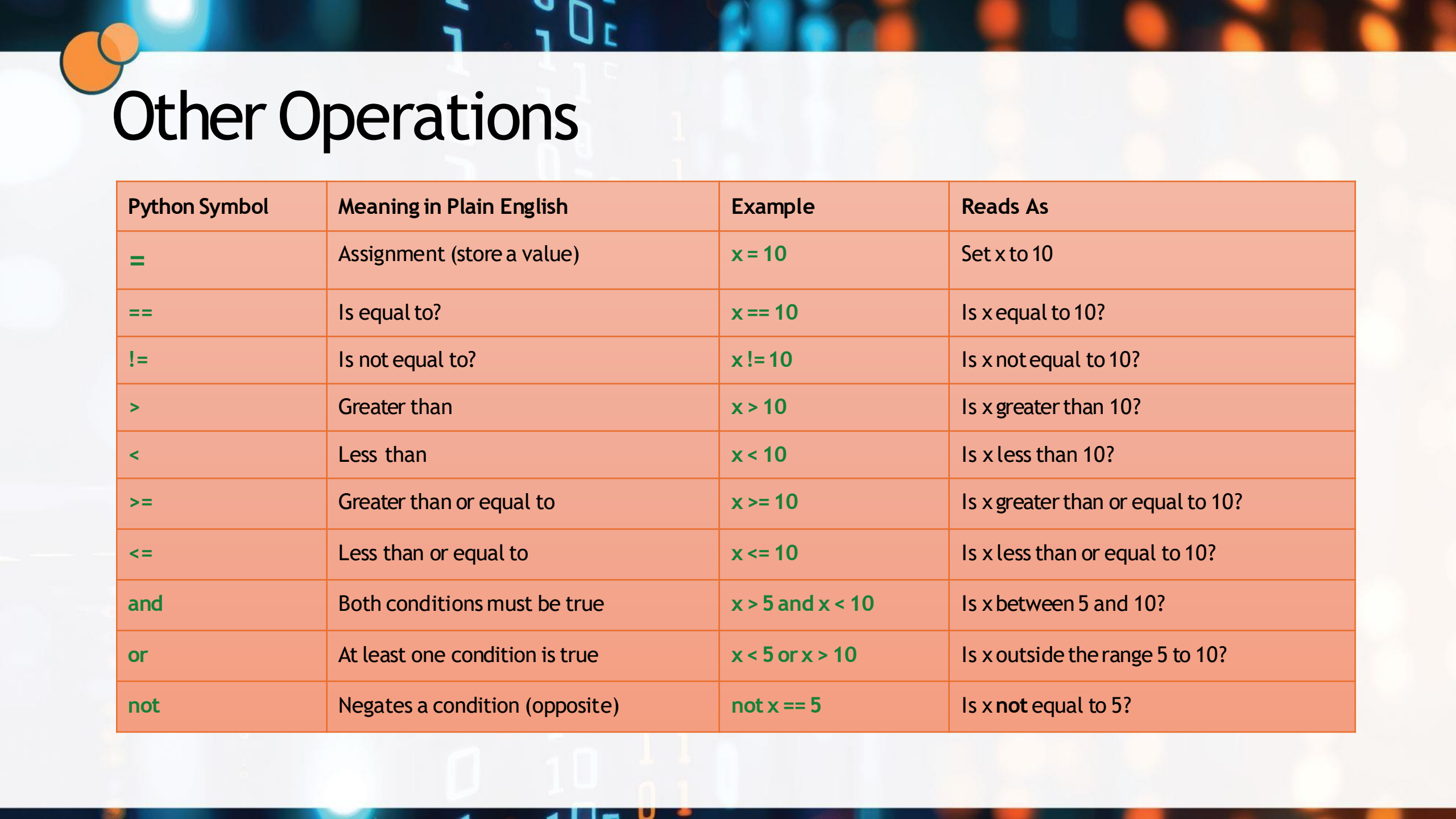
`type(variablename)`





Arithmetic Operations

Python Symbol	Math Symbol	Operation	Example
+	+	Addition	15 + 2
-	-	Subtraction	15 - 2
*	X	Multiplication	15 * 2
/	÷	Division	15 / 2
**	4^2	Exponent / Power	15 ** 2
()	()	Parentheses (brackets/grouping)	(15+2)*2



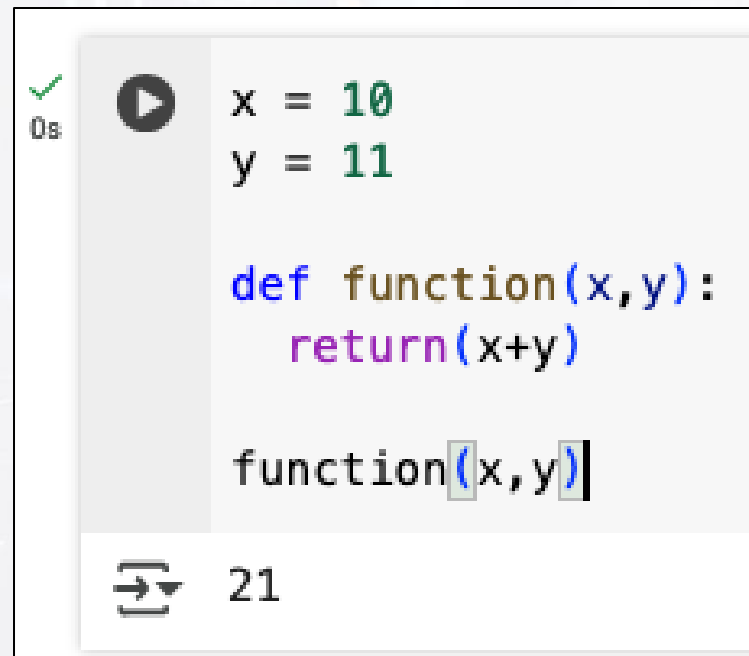
Other Operations

Python Symbol	Meaning in Plain English	Example	Reads As
<code>=</code>	Assignment (store a value)	<code>x = 10</code>	Set x to 10
<code>==</code>	Is equal to?	<code>x == 10</code>	Is x equal to 10?
<code>!=</code>	Is not equal to?	<code>x != 10</code>	Is x not equal to 10?
<code>></code>	Greater than	<code>x > 10</code>	Is x greater than 10?
<code><</code>	Less than	<code>x < 10</code>	Is x less than 10?
<code>>=</code>	Greater than or equal to	<code>x >= 10</code>	Is x greater than or equal to 10?
<code><=</code>	Less than or equal to	<code>x <= 10</code>	Is x less than or equal to 10?
<code>and</code>	Both conditions must be true	<code>x > 5 and x < 10</code>	Is x between 5 and 10?
<code>or</code>	At least one condition is true	<code>x < 5 or x > 10</code>	Is x outside the range 5 to 10?
<code>not</code>	Negates a condition (opposite)	<code>not x == 5</code>	Is x not equal to 5?



Functions

- To declare a function, use the def key word.



A screenshot of a Python REPL (Read-Eval-Print Loop) window. The window has a light gray background. On the left side, there is a vertical bar with a green checkmark and the text '0s'. Below this, there is a play button icon. The main area of the window contains the following code: `x = 10`, `y = 11`, `def function(x,y):`, `return(x+y)`, and `function(x,y)`. The cursor is positioned at the end of the last line. Below the code, there is a white box containing the output '21' and a return icon (two arrows pointing outwards).

```
x = 10
y = 11

def function(x,y):
    return(x+y)

function(x,y)
```

21



IF Statements

"If it's raining, take an umbrella. Else I will wear shorts"

This is an if statement in real life!





IF Statement

```
weather = "rainy"

if weather == "rainy":
    print("Take an umbrella.")
else:
    print("Take Shorts and Sunglasses")
```



Advance ELIF Statement

```
import random

# Step 1: Create a list (array) of weather conditions
weather_conditions = ["rainy", "sunny", "cloudy"]

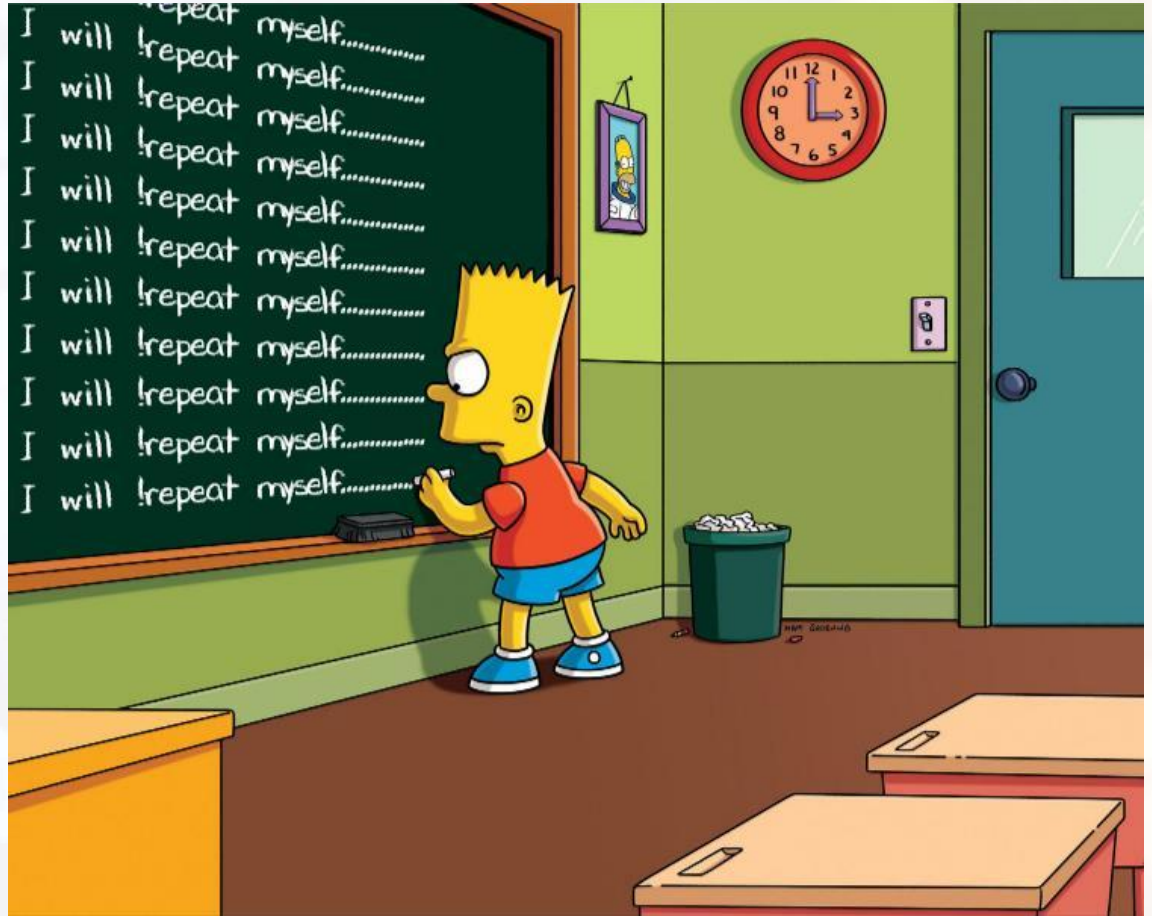
# Step 2: Randomly select a weather condition
weather = random.choice(weather_conditions)

print(f'Today\'s weather is {weather} we should...')

if weather == "rainy":
    print("Take an umbrella.")
elif weather == "sunny" or weather == "hot":
    print("Take shorts and sunglasses.")
elif weather == "cloudy":
    print("You might want a light jacket.")
```


Loops

- A loop is a sequence of instructions that is continually repeated until a certain condition is reached.
- *A loop is like telling the computer: “Keep doing this until I say stop”*





While Loop

- Print a countdown from 10 to 0

Print a countdown from 10 to 0

```
print(10)
print(9)
print(8)
print(7)
print(6)
print(5)
print(4)
print(3)
print(2)
print(1)
print(0)
print("🚀 Blast off!")
```



```
i = 10
while i > 0:
    print("T-Minus: ", i)
    i = i - 1

print("🚀 Blast off!")
```



While Loop

Print a countdown from 10 to 0

```
i = 10
```

```
while i >= 0:
```

```
    print("T-Minus: ", i)
```

```
    i = i - 1
```

```
print("🚀 Blast off!")
```



For Loops

Count from 0 till 10

```
for i in range(0, 10):  
    print("Number:", i)
```



For Loops

Count from 0 till 10

```
for i in range(0, 11):  
    print("Number:", i)
```




For Loops - Steps

count from 0 till 10 in steps of 2

```
for i in range(0, 12, 2):  
    print(f"We are now at {i}")
```



List/Arrays

- A list(array) is a special variable that can hold more than one value at a time.

	Start Index		End Index		
Index	0	1	2	3	4
Value	"Cats"	"Dogs"	"Rabbits"	"Gerbils"	"Hamsters"
	1st Element	2nd Element	3rd Element	4th Element	5th Element
Length = 5					



Creating and accessing List element

- These elements are separated by a comma and enclosed with square brackets.

Creating an array

```
proudly_sa = ['Bathu', 'K-Way', 'Drip', 'Maxhosa', 'Drip']
```

```
print(proudly_sa[0]) #Bathu
```

```
print(proudly_sa[3]) #Maxhosa
```



Accessing All Items using For Loop

- Could simply type 'proudly_sa' to see all elements or use For Loop:

Accessing elements by index

```
proudly_sa = ['Bathu', 'K-Way', 'Drip', 'Maxhosa', 'Drip']
```

```
for brand in proudly_sa:  
    print("Proudly South African brand:", brand)
```



Modifying a List

```
proudly_sa = ['Bathu', 'K-Way', 'Drip', 'Maxhosa', 'Drip']

proudly_sa[2] = "Drip Sportif" # Modify Drip to Drip Sportif
print(proudly_sa)

proudly_sa.remove("K-Way") # Only removes the first "K-Way"
print(proudly_sa)

proudly_sa.append("Tshepo Jeans") # add a new brand
print(proudly_sa)
```




More list functions

proudly_sa.	
Proudly Sou	 remove
Proudly Sou	 append
Proudly Sou	 clear
Proudly Sou	 copy
Proudly Sou	 count
	 extend
	 index
	 insert
	 pop
	 reverse
	 sort



Dictionary

- A Dictionary in Python is a collection of key-value pairs used to store data values.
- To declare a dictionary:

```
✓ [6] dict = {  
0s      1: 'python',  
        2: 'example',  
        3: 'data',  
        }  
  
✓ dict[1]  
0s      'python'
```

```
return 0;
```

n ICIS

NATIONAL INTEGRATED
CYBERINFRASTRUCTURE SYSTEM

DIRISA